

Kyle Main

Dallas/Ft. Worth Area | 469-545-3791 | main.kyle87@gmail.com

Machine Learning Engineer — Detection & Behavioral Analytics

Machine learning engineer and data scientist with 12 years of experience building systems that tell malicious behavior apart from normal behavior at scale — including 8+ years of hands-on ML detection engineering (unsupervised clustering models, time-series anomaly detection, and precision-tuned feature engineering) across massive telemetry volumes. Full-lifecycle ML experience: exploratory data analysis at scale (PySpark/SparkSQL on GCP Dataproc), model development with pandas/scikit-learn/PyTorch, and production deployment of detection content with continuous false-negative/false-positive analysis feeding back into feature engineering — the same efficacy loop central to this role.

CORE SKILLS

ML & Data Science: Python, SQL, pandas, scikit-learn, NumPy, SciPy, PyTorch, clustering/unsupervised ML, time-series anomaly detection, R

Data Analytics at Scale: PySpark / SparkSQL, GCP Dataproc, Zeppelin notebooks, BigQuery, Apache Beam / GCP Dataflow

Detection & Feature Engineering: Signature, statistical, behavioral, and ML-based detection rule development; entity/behavior baseline modeling; false-negative/false-positive analysis and tuning across attack categories

Production ML Pipelines: Detection-as-code CI/CD (GitLab), automated testing, rule-quality metrics tracking, staged/safe production rollout

Engineering: Git, Docker, structured/tested Python code, Elasticsearch

PROFESSIONAL EXPERIENCE

Senior Cybersecurity Engineer — Shorepoint

Oct 2023 – Present

- Built a UEBA (User and Entity Behavior Analytics) detection layer on custom Elasticsearch data transforms, modeling enterprise-wide behavioral baselines from CrowdStrike, Suricata, and Zeek telemetry to surface deviations from normal activity (DOE/NNSA Security Data Integration project, completed).
- Currently build and tune production detection analytics for Treasury's SOC, translating false-positive/false-negative investigation findings directly into refined detection logic (current project, Treasury SOC).

Senior Threat Detection Engineer and Data Scientist — Forescout

Aug 2022 – Oct 2023

- Senior member of a data science and detection engineering team building signature, behavioral, statistical, time-series, and ML-based detection content against massive-scale customer data on a cloud-based big-data platform.

Threat Detection Engineer and Data Scientist — Trend Micro / Cysiv

Sep 2018 – Aug 2022

- Built ML-based detection content clustering devices on the network by behavioral features (unsupervised learning) to surface anomalies signature-based rules missed — part of 2,300+ detection rules covering the MITRE ATT&CK matrix.
- Designed time-series anomaly detection models for entity behaviors — Outlook process chains, parent/child process combinations, and authentication patterns (volume and geography) — to flag deviation from an established baseline, directly analogous to modeling communication-pattern baselines and combining discriminative signals into a precise detection system.
- Ran exploratory data analysis at scale on GCP Dataproc compute clusters via Zeppelin notebooks and PySpark/SparkSQL, building a reusable analysis toolkit to validate new detection signal candidates before productionizing them.
- Owned data engineering feeding the detection pipeline — 220+ ingested sources, 50+ Logstash filters, and a Common Information Model standardizing fields across all of it — so model inputs stayed reliable at production scale.

Security Data Scientist — Experian

Jan 2015 – Jan 2018

- Analyzed large-scale security datasets to build custom models and algorithms identifying emerging threats — DNS-based malware detection/mitigation and anomalous-behavior discovery in log data.

EDUCATION & CERTIFICATIONS

M.S. Physics — University of North Texas (2013) | B.S. Physics — Ball State University (2011)

Splunk User Certification · Splunk for Analytics and Data Science · Johns Hopkins (Coursera): R Programming, The Data Scientist's Toolbox

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August 1, 2026

Hiring Team

Abnormal Security

Abnormal AI's Attack Detection team is solving a problem I've spent 12 years on from a different angle — telling malicious behavior apart from normal behavior at massive scale, in near-real time, using models and rules built to catch what static signatures miss.

At Trend Micro/Cysiv, I built machine-learning-based detection content that clustered devices on the network by behavioral features to flag anomalies invisible to signature-based rules, and designed time-series anomaly detection models against entity behaviors — Outlook process chains, parent/child process relationships, and authentication patterns across volume and geography — to catch deviation from an established baseline. That's the same modeling problem your team runs against communication patterns: building discriminative signals at the message, sender, and recipient level and combining them into a precise detection system.

I ran exploratory data analysis at scale on GCP Dataproc clusters using PySpark and SparkSQL to find and validate new detection signal candidates, then took those signals into production as part of a rule set covering 2,300+ detection use cases across the MITRE ATT&CK matrix — running the same efficacy loop this role calls for: analyzing false-negative and false-positive datasets, feeding findings back into feature engineering and model tuning, and monitoring production detection rates against a shifting attack landscape. I'm fluent in Python and the core ML toolkit — pandas, scikit-learn, NumPy, PyTorch — and comfortable taking a well-defined dataset through training, evaluation, and production deployment.

I'd welcome the chance to talk through how that background applies to catching the next attack pattern your Detection Engine hasn't seen yet.

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